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Control device for controlling multiple operating characteristics of an electrical load

Abstract

A load control device may be configured to control multiple characteristics of one or more electrical loads such as the intensity and color of a lighting load. The load control device may switch from controlling one characteristic of the electrical loads to controlling another characteristic of the electrical loads based on the position and/or orientation of one or more components of the load control device. Such a position and/or orientation may be manipulated by moving the one or more components relative to a base portion of the load control device. The load control device may be a wall-mounted device or a battery-powered remote control device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/612,027, filed Jun. 2, 2017, which claims the benefit of U.S. Provisional Patent Application No. 62/345,449, filed Jun. 3, 2016, and U.S. Provisional Patent Application No. 62/356,288, filed Jun. 29, 2016, the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

(1) The intensity and/or color of a lighting fixture may be manipulated for a variety of purposes such as presentation, comfort, and well-being. Typical color control techniques may include dim to warm, correlated color temperature (CCT), and full color tuning. Dim to warm is a technique for adjusting the color temperature of a light source in proportion to the intensity so as to mimic the color shift of incandescent lamps with respect to intensity (e.g., warmer color temperature at lower light levels, and cooler color temperature at higher light levels). CCT-based color tuning is a technique for controlling the color temperature and intensity of a light source independently within specified parameters. Full color tuning is a technique for changing the emitted color spectrum of a light source by mixing several base colors (e.g., red, green, blue) in different proportions.

(2) Different types of intensity control/color tuning techniques may require different types of user interfaces. Dim to warm color control, for example, generally requires one control input. The input may be translated (e.g., by an LED driver) into an appropriate intensity and/or color control signal for driving the lighting fixture. CCT-based color tuning and/or full color control may require one control input for color temperature and a separate control input for light intensity. A user interface may also need to be capable of activating and/or deactivating preconfigured environmental settings (e.g., a lighting scene created by tuning one or more lighting fixtures to particular color and/or intensity values). Further, a user may desire to not only apply the aforementioned types of control, but also receive feedback about the type, amount and/or result of the control being applied.

(3) A traditional control device is usually designed to control one specific aspect of an electrical load. For example, a dimmer switch may be only capable of controlling the intensity of a lighting load. To control the color of the lighting load, a separate control device is often required. Using

multiple traditional control devices to control an electrical load not only increases the cost and work required to set up the concerned load control system, but also negatively affects the usability and aesthetic appeal of the system. The traditional control devices also have very simplistic user interfaces that lack a feedback mechanism for keeping a user informed about the type and/or amount of control being applied.

SUMMARY

(4) As described herein, a control device may be configured for use in a load control system to control respective amount of power delivered to one or more electrical loads. The control device may be external to the one or more electrical loads, and may include a base portion, a rotating portion rotatable with respect to the base portion, an actuation portion defining a front surface, and a control circuit. The base portion may be configured to be mounted to an electrical wallbox (e.g., when the control device is a dimmer switch) or over an existing mechanical switch (e.g., when the control device is a retrofit remote control device). When configured as a dimmer device, the control device may further include a load control circuit adapted to be electrically coupled in series between an AC power source and the one or more electrical loads for controlling power delivered to the one or more electrical loads. When configured as a retrofit remote control device, the control device may be mounted over a toggle actuator of a mechanical switch that controls whether power is delivered to the one or more electrical loads.

(5) The actuation portion of the control device may be moveable along an axis perpendicular to the base portion of the control device when the actuation portion is pushed in toward the base portion or pulled away from the base portion. The control circuit may be configured to generate first control data for changing a first characteristic (e.g., an intensity) of the one or more electrical loads in response to rotations of the rotating portion when the front surface of the actuation portion is located in a first plane along the axis, and generate second control data for changing a second characteristic (e.g., a color) of the one or more electrical loads in response to rotations of the rotating portion when the front surface of the actuation portion is located in a second plane parallel to the first plane. The control device may comprise a communication circuit configured to transmit, e.g., to the one or more electrical loads, a first control signal associated with the first control data and a second control signal associated with the second control data.

(6) The control device may further include one or more visual indicators configured to be illuminated by one or more light sources. The one or more visual indicators may comprise a light bar. When illuminated, the light bar may provide feedback about the first and/or second characteristics of the one or more electrical loads.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts an example load control system that includes one or more example control devices.

(2) FIG. 2 is a perspective view of an example control device that may be deployed as a dimmer switch and/or a remote control device of the load control system illustrated in FIG. 1.

(3) FIGS. 3A and 3B show example perspective views of the remote control device of FIG. 2 illustrating how the positions and/or orientations of one or more components of a control unit of the remote control device may be changed relative to a base portion in order to put the remote control device into a certain control mode.

(4) FIG. 4A is a perspective view of an example remote control device that may be deployed as a remote control device of the load control system illustrated in FIG. 1 with a control unit detached from a base portion.

(5) FIG. 4B are rear views of the control unit and the base portion of the remote control device

depicted in FIG. 4A.

(6) FIG. 4C is a front exploded view of the control unit for the remote control device depicted in FIG. 4A.

(7) FIG. 4D shows a rear exploded view of the control unit for the example remote control device depicted in FIG. 4C.

(8) FIG. 5 shows a simplified block diagram of an example control device that may be deployed as a remote control device of the load control system illustrated in FIG. 1.

(9) FIG. 6 shows a simplified block diagram of an example control device that may be deployed as a load control device (e.g., a dimmer switch) of the load control system illustrated in FIG. 1.

(10) FIG. 7 shows a simplified flowchart of an example control procedure that may be executed by a control device for controlling multiple characteristics of an electrical load.

DETAILED DESCRIPTION

(11) FIG. 1 is a simplified diagram of an example load control system. As shown, the load control system is configured as a lighting control system **100** for control of one or more lighting loads, such as a lighting load **102** that is installed in a ceiling-mounted downlight fixture **103** and a controllable lighting load **104** that is installed in a table lamp **105**. The lighting loads **102**, **104** shown in FIG. 1 may include light sources of different types (e.g., incandescent lamps, fluorescent lamps, and/or LED light sources). The lighting loads may have advanced features. For example, the lighting loads may be controlled to emit light of varying intensities and/or colors in response to a user command. The amount of power delivered to the lighting loads may be adjusted to an absolute level or by a relative amount. The lighting control system **100** may be configured to control one or more of the lighting loads (e.g., and/or other electrical loads) according to one or more configurable presets or scenes. These presets or scenes may correspond to, for example, predefined light intensities and/or colors, predefined entertainment settings such as music selection and/or volume settings, predefined window treatment settings such as positions of shades, predefined environmental settings such as HVAC settings, or any combination thereof. The presets or scenes may correspond to one or more specific electrical loads (e.g., bedside lamps, ceiling lights, etc.) and/or one or more specific locations (e.g., a room, an entire house, etc.).

(12) The lighting load **102** may be an example of a lighting load that is wired into a power control and/or delivery path of the lighting control system **100**. As such, the lighting load **102** may be controllable by a wall-mounted control device such as a dimmer switch. The lighting load **104** may be an example of a lighting load that is equipped with integral load control circuitry and/or wireless communication capabilities such that the lighting load may be controlled via a wireless control mechanism (e.g., by a remote control device).

(13) The lighting control system **100** may include one or more control devices for controlling the lighting loads **102**, **104** (e.g., controlling an amount of power delivered to the lighting loads). The lighting loads **102**, **104** may be controlled substantially in unison, or be controlled individually. For example, the lighting loads may be zoned so that the lighting load **102** may be controlled by a first control device, while the lighting load **104** may be controlled by a second control device. The control devices may be configured to turn the lighting loads **102**, **104** on and off. The control devices may be configured to control the magnitude of a load current conducted through the lighting loads (e.g., so as to control an intensity of the lighting loads **102**, **104** between a low-end intensity L.sub.LE and a high-end intensity L.sub.HE). The control devices may be configured to control an amount of power delivered to the lighting loads to an absolute level (e.g., to a maximum allowable amount), or by a relative amount (e.g., an increase of 10% from a current level). The control devices may be configured to control a color of the lighting load **102**, **104** (e.g., by controlling a color temperature of the lighting loads or by applying full color control over the lighting loads).

(14) The control devices may be configured to activate a preset associated with the lighting load **102**, **104** (e.g., a preset may be associated with one or more predetermined settings of the lighting

loads such as an intensity level of the lighting loads and/or a color of the lighting loads). The presets may be configured via the control device and/or via an external device (e.g., a mobile device) by way of a wireless communication circuit of the control device. The control devices may be configured to activate control of a zone. A zone may correspond to one or more electrical loads that are configured to be controlled by the control devices. A zone may be associated with a specific location (e.g., a living room) or multiple locations (e.g., an entire house with multiple rooms and hallways). The control devices may be configured to switch between different operational modes. An operational mode may be associated with controlling different types of electrical loads or different operational aspects of one or more electrical loads. Examples of operational modes may include a lighting control mode for controlling one or more lighting loads (e.g., which in turn may include a color control mode and an intensity control mode), an entertainment system control mode (e.g., for controlling music selection and/or the volume of an audio system), an HVAC system control mode, a winter treatment device control mode (e.g., for controlling one or more shades), and/or the like.

(15) The control device described herein may be, for example, a dimmer switch **110**, a retrofit remote control device **112**, a wall-mounted control device **114**, a tabletop remote control device **116**, and/or a handheld remote control device **118**, as shown in FIG. 1. The dimmer switch **110** may include a base portion (e.g., such as one or more of a yoke, a bezel, and an enclosure that may house electrical circuitry and or mechanical complements of the dimmer switch **110**) that is configured to be mounted to a standard electrical wallbox. Once mounted, the dimmer switch **110** may be coupled in series electrical connection between an alternating-current (AC) power source **105** and a lighting load that is wired into the control path of the dimmer switch **110** (e.g., such as the lighting load **102**). The dimmer switch **110** may receive an AC mains line voltage $V_{sub.AC}$ from the AC power source **105**, and may generate a control signal for controlling the lighting load **102**. The control signal may be generated via various phase-control techniques (e.g., a forward phase-control dimming technique or a reverse phase-control dimming technique). The dimmer switch **110** may be configured to receive wireless signals (e.g., from a remote control device) representative of commands to control the lighting load **102**, and generate respective control signals for executing the commands. Examples of wall-mounted dimmer switches are described in greater detail with reference to FIG. 13, and in commonly-assigned U.S. Pat. No. 7,242,150, issued Jul. 10, 2007, entitled DIMMER HAVING A POWER SUPPLY MONITORING CIRCUIT; U.S. Pat. No. 7,546,473, issued Jun. 9, 2009, entitled DIMMER HAVING A MICROPROCESSOR-CONTROLLED POWER SUPPLY; and U.S. Pat. No. 8,664,881, issued Mar. 4, 2014, entitled TWO-WIRE DIMMER SWITCH FOR LOW-POWER LOADS, the entire disclosures of which are hereby incorporated by reference.

(16) The retrofit remote control device **112** may be configured to be mounted to a mechanical switch (e.g., a toggle switch **122**, a paddle switch, a pushbutton switch, and/or other suitable switch) that may be pre-existing in the lighting control system **100**. Such a retrofit solution may provide energy savings and/or advanced control features, for example without requiring significant electrical re-wiring and/or without requiring the replacement of existing mechanical switches. As an example, a consumer may replace an existing lamp with the controllable lighting load **104**, switch a toggle switch **122** that is coupled to the lighting load **104** to the on position, install (e.g., mount) the remote control device **112** onto the toggle switch **122**, and associate the remote control device **112** with the lighting source **104**. The retrofit remoted control **112** may then be used to perform advanced functions that the toggle switch **122** may be incapable of performing (e.g., such as dimming the intensity level of the light output, changing the color of the light output, providing feedback to a user, etc.). As shown, the toggle switch **122** is coupled (e.g., via a series electrical connection) between the AC power source **105** and an electrical receptacle **120** into which the lighting load **104** may be plugged (e.g., as shown in FIG. 1). Alternative, the toggle switch **122** may be coupled between the AC power source **105** and one or more of the lighting loads **102**, **104**,

without the electrical receptacle **120**.

(17) The wall-mounted remote control device **114** may be configured to be mounted to a standard electrical wallbox and be electrically connected to the AC power source **105** for receiving power. The wall-mounted remote control device **114** may be configured to receive a user input and may generate and transmit a control signal (e.g., control data such as a digital message) for controlling the lighting loads **102**, **104** in response to the user input. The tabletop remote control device **116** may be configured to be placed on a surface (e.g., an end table or night stand), and may be powered by a direct-current (DC) power source (e.g., a battery or an external DC power supply plugged into an electrical outlet). The tabletop remote control device **116** may be configured to receive a user input, and may generate and transmit a signal (e.g., a digital message) for controlling the lighting loads **102**, **104** in response to the user input. The handheld remote control device **118** may be sized to fit into a user's hand, and may be powered by a direct-current (DC) power source (e.g., a battery or an external DC power supply plugged into an electrical outlet). The handheld remote control device **118** may be configured to receive a user input, and may generate and transmit a signal (e.g., a digital message) for controlling the lighting loads **102**, **104** in response to the user input. Examples of battery-powered remote controls are described in greater detail in commonly assigned U.S. Pat. No. 8,330,638, issued Dec. 11, 2012, entitled WIRELESS BATTERY POWERED REMOTE CONTROL HAVING MULTIPLE MOUNTING MEANS, and U.S. Pat. No. 7,573,208, issued Aug. 11, 2009, entitled METHOD OF PROGRAMMING A LIGHTING PRESET FROM A RADIO-FREQUENCY REMOTE CONTROL, the entire disclosures of which are hereby incorporated by reference.

(18) The control devices described herein (e.g., the dimmer switch **110** and/or remote control devices **112-118**) may each include a user input unit. The user input unit may be configured to receive (e.g., detect) user inputs for controlling one or more of the lighting loads **102**, **104**, and/or the control device itself. A plurality of mechanisms for receiving the user inputs may be implemented on the user input unit, including, for example, a rotating mechanism (e.g., such as a rotary knob or a dial), a button or switch or an imitation thereof, and a touch sensitive device (e.g., such as a capacitive touch surface) configured to detect both point actuations and gestures.

(19) The control devices described herein (e.g., the dimmer switch **110** and/or remote control devices **112-118**) may each include one or more visual indicators (e.g., a light bar) configured to be illuminated by one or more light sources (e.g., one or more LEDs). The one or more visual indicators may be provided on the user input unit or may be separate from the user input unit. The one or more visual indicators may be operable to provide feedback to a user of the control device. Such feedback may indicate, for example, a status of a lighting load (e.g., the lighting loads **102**, **104**) controlled by the control device. The status may reflect, for example, whether the lighting load is on or off, a present intensity of the lighting load, a color of the lighting load, and so on. The feedback may indicate a status of the control device itself, for example, such as a present operational mode of the control device (e.g., an intensity control mode or a color control mode), a power status of the control device (e.g., remaining battery power), and so on. As an example, the control device may provide feedback via the visual indicators while the control device is being actuated and/or after the control device is actuated. The feedback may indicate to the user that the control device is transmitting control signals (e.g., RF signals) in response to the actuation. The control device may be configured to keep the visual indicators illuminated while the condition triggering the feedback continues to exist. The control device may be configured to illuminate the visual indicators for a few seconds (e.g., 1-2 seconds) and then turn off the visual indicators (e.g., to conserve battery life).

(20) The control devices described herein (e.g., the dimmer switch **110** and/or remote control devices **112-118**) may each include a control circuit. The control circuit may be configured to be responsive to a user input received via the user input unit. The control circuit may be configured to generate control data (e.g., a control signal) for controlling the lighting loads **102**, **104** in response

to the user input. The control data may include commands and/or other information (e.g., device identification information) for controlling the lighting loads **102, 104**. The control data may be included in a control signal transmitted to the lighting loads **102, 104** via a wireless communication circuit. The control circuit may be configured to illuminate the one or more visual indicators to provide feedback of the control being applied and/or its outcome.

(21) The control devices described herein (e.g., the dimmer switch **110** and/or remote control devices **112-118**) may each include a wireless communication circuit for transmitting and/or receiving radio frequency (RF) signals **108**. The wireless communication circuit may be used to transmit a control signal that includes the control data (e.g., a digital message) generated by the control device to the lighting loads **102, 104** or to a central controller of the lighting control system **100**, for example. The control data may be generated in response to a user input to adjust one or more operational aspects of the lighting loads **102, 104**. The control data may include a command and/or identification information (e.g., such as a unique identifier) associated with the control device and/or one or more of the lighting loads **102, 104** (e.g., and/or other electrical loads of the load control system **100**).

(22) The control devices (e.g., the remote control devices **112-118**) may be associated with one or more lighting loads and/or other control devices (e.g., the dimmer switch **110**) for controlling the lighting loads (e.g., through a configuration procedure). Upon such association, the lighting loads **102, 104** may be responsive to control signals transmitted by the control devices. To illustrate, the association may be accomplished by actuating an actuator on the concerned lighting loads and/or control devices, and then actuating (e.g., pressing and holding) an actuator on the control device for a predetermined amount of time (e.g., approximately 10 seconds). Examples of a configuration procedure for associating a control device with an electrical load is described in greater detail in commonly-assigned U.S. Patent Publication No. 2008/0111491, published May 15, 2008, entitled RADIO-FREQUENCY LIGHTING CONTROL SYSTEM, the entire disclosure of which is hereby incorporated by reference. The wireless communication circuit may also be controlled to transmit/receive feedback information regarding the control device and/or the lighting loads **102, 104** via RF signals.

(23) The control device described herein (e.g., the dimmer switch **110** and/or remote control devices **112-118**) may include a memory (not shown). The memory may be used, for example, to store operational settings associated with the control device and/or the lighting loads **102, 104** (e.g., such as lighting presets and their associated light intensities and/or colors). The memory may be implemented as an external integrated circuit (IC) or as an internal circuit (e.g., as part of a control circuit).

(24) Further, it should be appreciated that, although a lighting control system with two lighting loads is provided as an example above, a load control system as described herein may include more or fewer lighting loads, other types of lighting loads, and/or other types of electrical loads that may be configured to be controlled by the one or more control devices. For example, the load control system may include one or more of: a dimming ballast for driving a gas-discharge lamp; an LED driver for driving an LED light source; a dimming circuit for controlling the intensity of a lighting load; a screw-in luminaire including a dimmer circuit and an incandescent or halogen lamp; a screw-in luminaire including a ballast and a compact fluorescent lamp; a screw-in luminaire including an LED driver and an LED light source; an electronic switch, controllable circuit breaker, or other switching device for turning an appliance on and off; a plug-in control device, controllable electrical receptacle, or controllable power strip for controlling one or more plug-in loads; a motor control unit for controlling a motor load, such as a ceiling fan or an exhaust fan; a drive unit for controlling a motorized window treatment or a projection screen; one or more motorized interior and/or exterior shutters; a thermostat for a heating and/or cooling system; a temperature control device for controlling a setpoint temperature of a heating, ventilation, and air-conditioning (HVAC) system; an air conditioner; a compressor; an electric baseboard heater controller; a controllable

damper; a variable air volume controller; a fresh air intake controller; a ventilation controller; one or more hydraulic valves for use in radiators and radiant heating system; a humidity control unit; a humidifier; a dehumidifier; a water heater; a boiler controller; a pool pump; a refrigerator; a freezer; a television and/or computer monitor; a video camera; an audio system or amplifier; an elevator; a power supply; a generator; an electric charger, such as an electric vehicle charger; an alternative energy controller; and/or the like.

(25) FIG. 2 depicts an example control device **200** that may be deployed as the dimmer switch **110** and/or the retrofit remote control device **112** in the lighting control system **100**. The lighting control system **100** may include one or more lighting loads, such as the lighting loads **102**, **104**. The control device **200** may comprise a user interface **210** (e.g., a user input device) and a faceplate **212**. The user interface **202** may include a rotating portion **214** that is rotatable with respect to the faceplate **212** for controlling multiple characteristics (e.g., two different operating characteristics) of the lighting loads controlled by the control device (e.g., adjusting the intensities and/or the colors of the lighting loads). The user interface **210** may also include an actuation portion **216** that may be pressed in towards the faceplate **212** for turning the lighting loads on and off (e.g., toggling the lighting loads). More generally, the control device **200** may be responsive to a dynamic motion of the actuation portion **216** (e.g., an actuation that causes movement of the surface of the actuation portion). The user interface **210** may also include one or more visual indicators (e.g., a light bar **218**) configured to be illuminated by one or more light sources (e.g., one or more LEDs) to visibly display information. The light bar **218** may be attached to a periphery of the actuation portion **216** and may move with the actuation portion **216** (e.g., when the actuation portion is actuated). The control device **200** may comprise a base portion **220** for rotatably supporting the rotation portion **214**.

(26) The positions and/or orientations of one or more components of the control device **200** may be changed relative to the faceplate **212** and/or the base portion **220** in order to put the control device **200** into a certain control mode as will be described in greater detail below. For example, the control device **200** (e.g., one or more components of the control device **200**) may be positioned in different planes along an axis substantially perpendicular to a plane of the front surface of the faceplate **206** and/or a plane of the base portion **220**. In an example, the rotating portion **214** may be pushed in towards and/or pulled away from the faceplate **212** and/or the base portion **220** to switch the control device **200** between an intensity control mode and a color control mode (e.g., a color temperature control mode).

(27) The rotating portion **214** may be connected to the actuation portion **216** and may move with the actuation portion **216** (e.g., move towards the faceplate **212** along an axis perpendicular to the faceplate) to actuate an internal switch (not shown) when the actuation portion **216** is actuated. The actuation portion **216** may have a front surface that may define a front area of the rotating portion **214**. In an alternative example implementation, the actuation portion **216** may be separate from the rotating portion **214** and may be received in a central circular opening defined by the rotating portion **214** to allow the actuation portion **216** to move in towards the faceplate **212** along the axis perpendicular to the faceplate to actuate the internal switch when the actuation portion **216** is actuated. The actuation portion **216** may return (e.g., move away from the faceplate **212** along the axis perpendicular to the faceplate) to an idle position (e.g., as shown in FIG. 2) after being actuated.

(28) When the actuation portion **216** is in the idle position, the front surface of the actuation portion **216** (e.g., the front area of the rotation portion **214**) may be located in a first plane that may be parallel to a front surface of the faceplate **212** (e.g., along an axis perpendicular to the front surface of the faceplate **212** or to the base portion **220**). When the front surface of the actuation portion **216** is in the first plane, the actuation portion **216** may be configured to not actuate the internal switch. The rotating portion **214** may be rotated to adjust the amount of power delivered to the lighting loads to adjust the intensities of the lighting loads when the front surface of the actuation portion

216 is in the first plane. For example, when the rotating portion **214** is rotated to adjust the intensity of the lighting loads while the front surface of the actuation portion **216** is in the first plane, the light bar **218** may be illuminated (e.g., illuminated in a single color, such as white) to display feedback information regarding the present intensity of one or more of the lighting loads.

(29) As shown in FIG. 3A, the rotating portion **214** and/or the actuation portion **216** may be pushed in towards the faceplate **212** to cause the front surface of the actuation portion **216** to be in a second plane that is parallel to the front surface of the faceplate and/or to the base portion **220**. The second plane may be closer to the faceplate than the first plane. When the front surface of the actuation portion **216** is in the second plane, the actuation portion **216** may be actuating (e.g., constantly actuating) the internal switch. The rotating portion **214** may be rotated to adjust a color (e.g., a color temperature) of the lighting load when the front surface of the actuation portion **216** is in the second plane. The front surface of the actuation portion **216** may be maintained in the second plane without being held in the second plane (e.g., without an external force that may be applied through user interaction). Alternatively or additionally, the user may press and hold the rotating portion **214** and/or the actuation portion **216** of the control device **200** in (e.g., towards the faceplate **212**) in order to place the control device **200** in the color control mode. The rotating portion **214** and/or the actuation portion **216** may be held by an external force (e.g., which may be applied by a user) while the front surface of the actuation portion **216** is in the second plane. The rotating portion **214** and/or the actuation portion **216** may be biased (e.g., spring biased) back to return the front surface of the actuation portion **216** to the first plane when the external force is no longer applied (e.g., when the rotating portion **214** is released).

(30) In addition, as shown in FIG. 3B, the rotating portion **214** and/or the actuation portion **216** may be pulled out away from the faceplate **212** (e.g., along an axis perpendicular to the faceplate **212** and/or the base portion **216**) to cause the front surface of the actuation portion **216** to be in a third plane that is parallel to the front surface of the faceplate and farther away from the faceplate than the first plane. The control device **200** may be configured to detect that the front surface of the actuation portion **216** has been moved from the first plane to the third plane. For example, the control device **200** may include a switch that is configured to be open (e.g., non-conductive) when the front surface of the actuation portion **216** is in the first plane and closed (e.g., conductive) when the front surface of the actuation portion **216** is in the third plane. In response to a movement of the actuation portion **216**, the control circuit may receive an indication of the state of the switch (e.g., open or close) and determine the position of the actuation portion **216** accordingly. Alternatively or additionally, the control device **200** may be configured to detect the position of the actuation portion **216** via a sensor (e.g., a light emitting diode (LED) sensor), a photosensitive device, an optocoupler that comprises an infra-red (IR) light emitting diode (LED) and a photodiode, an inductive sensor, a hall-effect sensor circuit, a manually operated switch, and/or the like.

(31) When the front surface of the actuation portion **216** is in the third plane, the rotating portion **214** may be rotated to adjust the color (e.g., the color temperature) of the lighting load.

(32) The rotating portion **214** may be maintained in the third plane without being held in the third plane (e.g., without an external force applied through user interaction). Alternatively or additionally, the user may pull and hold rotating portion **214** and/or the actuation portion **216** of the control device **200** (e.g., away from the faceplate **212**) in order to place the control device **200** in the color control mode. The rotating portion **214** and/or the actuation portion **216** may be held by an external force (e.g., which may be applied by a user) while the front surface of the actuation portion **216** is in the third plane. The rotating portion **214** and/or the actuation portion **216** may be biased (e.g., spring biased) back to return the front surface of the actuation portion **216** to the first plane when the external force is no longer applied (e.g., when the rotating portion **214** is released).

(33) Slow rotations of the rotating portion **214** while the front surface of the actuation portion **216** is in the second plane or the third plane may provide fine tune adjustments of the color of the lighting loads. Fast rotations of the rotating portion **214** while the front surface of the actuation

portion **216** is in the second plane or the third plane may provide gross adjustments of the color of the lighting loads.

(34) When the rotation portion **214** is rotated to adjust the color of the lighting loads while the front surface of the actuation portion **216** is in the second plane or the third plane, the light bar **218** may be illuminated, for example, with one or more colors to provide feedback of the present color of one or more of the lighting loads. For example, the entire light bar **218** may be illuminated a single color to provide feedback of the present color of one or more of the lighting loads. Additionally or alternatively, the front surface of the actuation portion **216** may be illuminated a single color (e.g., from behind by LEDs inside of the control device **200**) to provide feedback of the present color of one or more of the lighting loads.

(35) Although the control device **200** is described herein as having two operational modes corresponding to different positions of the front surface of the actuation portion **216**, it will be appreciated that the control device **200** is not limited to having only two operational modes. The actuation portion **216** may be configured to be moveable among multiple positions (e.g., more than two positions) along the axis that is perpendicular to the faceplate **212**. Each of those positions may correspond to a different mode of the control device **200** (e.g., intensity, color temperature, full color, etc.). The actuation portion **216** may also be configured to have different orientations (e.g., tilted at different angles with respect to a plane of the faceplate **212**) that correspond to various operational modes of the control device **200**.

(36) FIGS. **4A** and **4B** are front and rear exploded perspective views of another example remote control device **310** that may be deployed as the retrofit remote control device **112** in the lighting control system **100** shown in FIG. **1** and/or the control device **200** shown in FIG. **2**. The remote control device **310** may be configured to be mounted over an actuator of a standard light switch **312** (e.g., a toggle actuator of a single pole single throw (SPST) maintained mechanical switch). The remote control device **310** may be installed over of an existing faceplate **316** that is mounted to the light switch **312** (e.g., via faceplate screws **318**). The remote control device **310** may include a base portion **320** and a control unit **330** that may be operably coupled to the base portion **320**. The control unit **330** may be supported by the base portion **320** and may include a rotating portion **332** (e.g., an annular rotating portion) that is rotatable with respect to the base portion **320**.

(37) As shown in FIG. **4A**, the control unit **330** may be detached from the base portion **320**. The base portion **320** may be attached (e.g., fixedly attached) to a toggle actuator **314** and may be configured to maintain the toggle actuator **314** in the on position. The toggle actuator **314** may be received through a toggle actuator opening **322** in the base portion **320**. A screw **324** may be tightened to attach (e.g., fixedly attached) the base portion **320** to the toggle actuator **314**. In this regard, the base portion **320** may be configured to prevent a user from inadvertently switching the toggle actuator **314** to the off position when the remote control device **310** is attached to the light switch **312**.

(38) The control unit **330** may be released from the base portion **320**. For example, a control unit release tab **326** may be provided on the base portion **320**. By actuating the control unit release tab **326** (e.g., pushing up towards the base portion or pulling down away from the base portion), a user may remove the control unit **330** from the base portion **320**.

(39) The control unit **330** may comprise one or more clips **338** that may be retained by respective locking members **328** connected to the control unit release tab **326** when the base portion **320** is in a locked position. The one or more clips **338** may be released from the respective locking members **328** of the base portion **320** when the control unit release tab **326** is actuated (e.g., pushed up towards the base portion or pulled down away from the base portion) to put the base portion **320** in an unlocked position. In an example, the locking members **328** may be spring biased into the locked position and may automatically return to the locked position after the control unit release tab **326** is actuated and released. In an example, the locking members **328** may not be spring biased, in which case the control unit release tab **326** may be actuated to return the base portion **320**

to the locked position.

(40) The control unit **330** may be installed on the base portion **320** without adjusting the base portion **320** to the unlocked position. For example, the one or more clips **338** of the control unit **330** may be configured to flex around the respective locking members **328** of the base portion and snap into place, such that the control unit **330** is fixedly attached to the base portion.

(41) The control unit **330** may be released from the base portion **320** to access one or more batteries **340** (e.g., as shown in FIG. 4B) that provides power to at least the remote control device **310**. The batteries **340** may be held in place in various ways. For example, the batteries **340** may be held by a battery retention strap **342**, which may also operate as an electrical contact for the batteries. The battery retention strap **342** may be loosened by untightening a battery retention screw **344** to allow the batteries **340** to be removed and replaced. Although FIG. 4B depicts the batteries **340** as being located in the control unit **330**, it should be appreciated that the batteries **340** may be placed elsewhere in the remote control device **310** (e.g., in the base portion **320**) without affecting the functionality of the remote control device **310**.

(42) When the control unit **330** is coupled to the base portion **320**, the rotating portion **332** may be rotatable in opposed directions about the base portion **320** (e.g., in the clockwise and/or counter-clockwise directions). The base portion **320** may be configured to be mounted over the toggle actuator **314** of the switch **312** such that the rotational movement of the rotating portion **332** may not change the operational state of the toggle actuator **314** (e.g., the toggle actuator **314** may remain in the on position to maintain functionality of the remote control device **310**).

(43) The control unit **330** may comprise an actuation portion **334**. The actuation portion **334** may in turn comprise a part or an entirety of a front surface of the control unit **330**. For example, the control unit **330** may have a circular surface within an opening defined by the rotating portion **332**. The actuation portion **334** may comprise a part of the circular surface (e.g., a central area of the circular surface) or approximately the entire circular surface. The actuation portion **334** may be received in a central circular opening defined by the rotating portion **332**. In an example, the actuation portion **334** may be configured to move towards the light switch **312** (e.g., through the central opening of the rotation portion **332**) to actuate a mechanical switch (not shown) inside the control unit **330** as will be described in greater detail below. The actuation portion **334** may return to an idle position after being actuated. In addition, the rotating portion **332** may be connected to the actuation portion **334** and may move with the actuation portion to actuate the mechanical switch when the actuation portion **332** is actuated.

(44) When the actuation portion **334** is in the idle position, a front surface of the actuation portion **334** may be located in a first plane that may be parallel to a front surface of the base portion **320**. The rotating portion **332** and/or the actuation portion **334** may be pushed into towards the base portion **320** to cause the front surface of the actuation portion **334** to be in a second plane that is parallel to the front surface of the faceplate and closer to the faceplate than the first plane. In addition, the rotating portion **332** and/or the actuation portion **334** may be pulled out away from the base portion **320** to cause the front surface of the actuation portion **334** to be in a third plane that is parallel to the front surface of the faceplate and farther away from the faceplate than the first plane.

(45) The remote control device **310** may be configured to transmit one or more wireless communication signals (e.g., the RF signals **108** of FIG. 1) to an electrical load (e.g., the lighting loads **102**, **104** of the lighting control system **100** of FIG. 1). The remote control device **310** may include a wireless communication circuit (e.g., an RF transceiver or transmitter (not shown)) via which one or more wireless communication signals may be sent and/or received. The control unit **330** may be configured to transmit digital messages (e.g., including commands to control the controllable electrical load) via the wireless communication signals. For example, when the front surface of the actuation portion **334** is in the first plane, the control unit **330** may be configured to transmit a command, via the wireless communication circuit, to raise the intensity of a controllable lighting load in response to a clockwise rotation of the rotating portion **332** and to transmit a

command to lower the intensity of the controllable light source in response to a counterclockwise rotation of the rotating portion **332**. In addition, when the front surface of the actuation portion **334** is in either the second plane or the third plane, the control unit **330** may be configured to transmit a command, via the wireless communication circuit, to adjust the color (e.g., the color temperature) of the controllable light source in response to clockwise and counterclockwise rotations of the rotating portion **332**.

(46) The control unit **330** may be configured to transmit a command to toggle an electrical load (e.g., from off to on or vice versa) in response to an actuation of the actuation portion **334**. In addition, the control unit **330** may be configured to transmit a command to turn an electrical load on in response to an actuation of the actuation portion **334** (e.g., if the control unit **330** possesses information indicating that the electrical load is presently off). The control unit **330** may be configured to transmit a command to turn an electrical load off in response to an actuation of the actuation portion **334** (e.g., if the control unit possesses information indicating that the electrical load is presently on).

(47) The control unit **330** may be configured to transmit a command to turn an electrical load on to a maximum power level (e.g., to turn a light source on to full intensity) in response to a double tap of the actuation portion **334** (e.g., two actuations in quick succession). The control unit **330** may be configured to adjust the power level of an electrical load to a minimum level (e.g., to turn the intensity of a lighting load to a minimum intensity) in response to rotation of the rotating portion **332** and may only turn off the electrical load in response to an actuation of the actuation portion **334**. The control unit **330** may also be configured in a spin-to-off mode, in which the control unit **330** may turn off an electrical load after the power level of the electrical load (e.g., intensity of the lighting load) is controlled to a minimum level in response to a rotation of the rotating portion **332** (e.g., without an actuation of the actuation portion).

(48) The control unit **330** may comprise one or more visual indicators (e.g., a light bar **336**) that may be illuminated by one or light sources (e.g., LEDs), for example, to provide feedback to a user of the remoted control device **310**. The light bar **336** may be located in different areas of the remote control device **310** in different implementations. For example, the light bar **336** may be located between the rotating portion **332** and the actuation portion **334**, and/or extend along the perimeter of the rotating portion **332** or the actuation portion **334**. The light bar **336** may have different shapes. For example, the light bar **336** may form a full circle (e.g., a substantially full circle) as shown in FIG. **4A**, a partial circle, a linear light bar, and/or the like. The light bar **336** may be attached to a periphery of the actuation portion **334** and move with the actuation portion **334** (e.g., when the actuation portion is actuated). The light bar **336** may have a certain width (e.g., a same width along the entire length of the light bar). The exact value of the width may vary, for example, depending on the size of the remote control device **310** and/or the intensity of the light source(s) that illuminates the light bar **336**.

(49) FIG. **4C** is a front exploded view and FIG. **4D** is a rear exploded view of the control unit **330** of the remote control device **310**. The actuation portion **334** may be received within an opening defined by the rotating portion **332**. The light bar **336** may be attached to the actuation portion **334** around a periphery of the actuation portion. The rotating portion **332** may comprise an inner surface **416** having tabs **418** surrounding the circumference of the rotation portion. The tabs **418** may be separated by notches **420** that are configured to receive engagement members **422** of the actuation portion **334** to thus engage the actuation portion **334** with the rotating portion **332**. The control unit **330** may also comprise a bushing **424** that is received within the rotating portion **332**, such that an upper surface **426** of the bushing may contact lower surfaces **428** of the tabs **418** inside of the rotating portion.

(50) When the actuation portion **334** is received within the opening of the rotating portion **332**, the light bar **336** may be provided between the actuation portion **334** and the rotating portion **332**. When the rotating portion **334** is rotated, the actuation portion **334** and/or the light bar **336** may

rotate with the rotating portion. The engagement members **422** of the actuation portion **334** may be able to move through the notches **420** in a z-direction (e.g., towards the base portion), such that the actuation portion **334** (along with the light bar **336**) may be able to move in the z-direction.

(51) The control unit **330** may further comprise a flexible printed circuit board (PCB) **430** that may be arranged over a carrier **432**. The flexible PCB **430** may comprise a main portion **434** on which most of the control circuitry of the control unit **330** (e.g., including a control circuit) may be mounted. The control unit **330** may comprise a plurality of light-emitting diodes (LEDs) **436** arranged around the perimeter of the flexible PCB **430** to illuminating the light bar **336**. The flexible PCB **430** may comprise a switch tab **438** that may be connected to the main portion **434** (e.g., via flexible arms **440**). The switch tab **438** may have a mechanical tactile switch **442** mounted thereto. The switch tab **438** of the flexible PCB **430** may be configured to rest on a switch tab surface **444** on the carrier **432**. The carrier **432** may comprise engagement members **446** configured to be received within notches **448** in the bushing **424**. A ring **450** may snap to a lower surface **452** of the rotating portion to hold the control unit **330** together. The clips **338** may be attached to the carrier **432** to allow the control unit **330** to be connected to the base portion.

(52) When the actuation portion **334** is pressed, the actuation portion **334** may move along the z-direction until an inner surface **458** of the actuation member actuates the mechanical tactile switch **442**. The actuation portion **334** may be returned to the idle position by the mechanical tactile switch **442**. In addition, the control unit **330** may comprise an additional return spring for returning the actuation portion **334** to the idle position. In some examples, actuations of the actuation portion **334** may not cause the actuation portion to move (e.g., the actuation portion **334** may substantially maintain its position along the z-direction). For example, the front surface of the actuation portion **334** may be a touch sensitive surface (e.g., a capacitive touch surface) configured to detect a user input via a point actuation and/or a gesture.

(53) The batteries **340** may be adapted to be received within a battery recess **462** in the carrier **432** as shown in FIG. **4D**. The batteries **340** may be held in place by the battery retention strap **342**, which may also operate as a negative electrical contact for the batteries and tamper resistant fastener for the batteries. The flexible PCB may comprise a contact pad **466** that may operate as a positive electrical contact for the batteries **340**. The battery retention strap **342** may comprise a leg **468** that ends in a foot **470** that may be electrically connected to a flexible pad **472** (e.g., as shown in FIG. **4C**) on the flexible PCB **430**. The battery retention strap **342** may be held in place by the battery retention screw **344** received in an opening **476** in the carrier **432**. When the battery retention screw **344** is loosened and removed from the opening **476**, the flexible pad **472** may be configured to move (e.g., bend or twist) to allow the battery retention strap **342** to move out of the way of the batteries **340** to allow the batteries to be removed and replaced.

(54) The control unit **330** may further comprise a magnetic strip **480** located on the inner surface **416** of the rotating portion **332** and extending around the circumference of the rotating portion. The flexible PCB **430** may comprise a rotational sensor pad **482** on which a rotational sensor (e.g., a Hall effect sensor integrated circuit **484**) may be mounted. The rotational sensor pad **482** may be arranged perpendicular to the main portion **434** of the flexible PCB **430** as shown in FIG. **4D**. The magnetic strip **480** may comprise a plurality of alternating positive and negative sections, and the Hall effect sensor integrated circuit **484** may comprise two sensor circuits operable to detect the passing of the positive and negative sections of the magnetic strip as the rotating portion **332** is rotated. Accordingly, the control circuit of the control unit **330** may be configured to determine the rotational speed and direction of rotation of the rotation portion **332** in response to the Hall effect sensor integrated circuit **484**. The flexible PCB **430** may also comprise a programming tab **486** to allow for programming of the control circuit of the control unit **330**.

(55) As shown in FIG. **4D**, the carrier **432** may comprise an actuator opening **490** adapted to receive the toggle actuator of the light switch when the control unit **330** is mounted to the base portion. The carrier **432** may comprise a flat portion **492** that may prevent the toggle actuator of the

light switch from extending into the inner structure of the control unit **330** (e.g., if the toggle actuator is particularly long). The flexible PCB **430** may also comprise an antenna **494** on an antenna tab **496** that may lay against the flat portion **492** in the actuator opening **490**.

(56) While the rotating portions **214**, **332** and the actuations portions **216**, **334** of the control device **200** and the remote control device **310** shown and described herein have a circular shape, the rotating portions and the actuation portions could have other shapes. For example, the rotating portions and the actuation portions may a rectangular shape, a square shape, a diamond shape, a triangular shape, an oval shape, a star shape, or any suitable shape. The front surface of the actuations portions **216**, **334** and/or the side surfaces of the rotating portions **214**, **332** may be planar or non-planar. In addition, the light bars **218**, **336** may have alternative shapes, such as a rectangular shape, a square shape, a diamond shape, a triangular shape, an oval shape, a star shape, or any suitable shape. The light bars **218**, **336** may be continuous loops, partial loops, broken loops, a single linear bar, a linear or circular array of visual indicators, and/or other suitable arrangement. The surfaces of the control device **200** and/or the remote control device **300** may be characterized by various colors, finishes, designs, patterns, etc.

(57) FIG. 5 is a simplified block diagram of an example control device **500** (e.g., a remote control device), which may be deployed as the remote control devices **112-118** in the lighting control system **100**, the control device **200**, and/or the remote control devices **310**. The control device **500** may include a control circuit **530**, one or more actuators **532** (e.g., buttons and/or switches), a rotational sensing circuit **534**, a wireless communication circuit **538**, a memory **540**, a battery **542**, and/or one or more LEDs **544**. The memory **540** may be configured to store one or more operating parameters (e.g., such as a preconfigured color scene or a preset light intensity) of the control device **500**. The battery **542** may provide power to one or more of the components shown in FIG. 5.

(58) The one or more actuators **532** may include a button or switch (e.g., a mechanical button or switch, or an imitation thereof) such as those described in association with the actuation portion **216** of the control device **200** and/or the actuation portion **334** of the remote control device **310**. The actuators **532** may be configured to send respective input signals to the control circuit **530** in response to actuations of the actuators **532** (e.g., in response to movements of the actuators **532**). The rotational sensing circuit **534** may be configured to translate a force applied to a rotating mechanism (e.g., such as the rotating portion **214** of the control device **200** and/or the rotating portion **334** of the remote control device **310**) into an input signal and provide the input signal to the control circuit **530**. The rotational sensing circuit **534** may include, for example, one or more magnetic sensors (such as Hall-effect sensors (HES), tunneling magnetoresistance (TMR) sensors, anisotropic magnetoresistance (AMR) sensors, giant magnetoresistance (GMR) sensors, reed switches, or other mechanical magnetic sensors), a mechanical encoder, an optical encoder, and/or a potentiometer (e.g., a polymer thick film or other resistive trace on a printed circuit board).

(59) The control circuit **530** may be configured to translate the input signals provided by the actuators **534** and/or the rotational sensing circuit **534** into control data (e.g., digital control signals) for controlling one or more electrical loads. The control circuit **530** may cause the control data (e.g., digital control signals) to be transmitted to the electrical loads via the wireless communication circuit **538**. For example, the wireless communication circuit **538** may transmit a control signal including the control data to the one or more electrical loads or to a central controller of the concerned load control system. The control circuit **530** may transmit a control signal including control data for turning one or more lighting loads on or off in response to an actuation of one of the actuators **534**. The control circuit **530** may transmit one or more control signals including control data for adjusting the intensities of one or more lighting loads in response to rotations of the rotating mechanism determined from the rotational sensing circuit **534**. The control circuit **530** may transmit one or more control signals including control data for adjusting the color (e.g., the color temperature) of one or more lighting loads in response to rotations of the rotating

mechanism while one of the actuators **534** is being actuated.

(60) The control circuit **530** may illuminate the LEDs **544** to present a light bar (e.g., such as the light bar **218** and/or the light bar **336**) and/or one or more indicator lights to provide feedback about various conditions. When the control circuit **530** is transmitting control signals including control data for adjusting the intensities of one or more lighting loads, the control circuit may control the LEDs **544** to illuminate the light bar (e.g., illuminated in a single color, such as white) to display feedback information regarding the present intensity of one or more of the lighting loads. When the control circuit **530** is transmitting control signals including control data for adjusting the color of one or more lighting loads, the control circuit may control the LEDs **544** to illuminate the light bar with one or more colors to provide feedback of the present color of one or more of the lighting loads.

(61) FIG. **6** is a simplified block diagram of an example control device **600** (e.g., a dimmer switch) that may be deployed as, for example, the dimmer switch **110** of the lighting control system **100** and/or the control device **200**. The control device **600** may include a hot terminal H that may be adapted to be coupled to an AC power source **602**. The control device **600** may include a controlled hot terminal CH (e.g., a switched hot and/or a dimmed hot terminal) that may be adapted to be coupled to an electrical load, such as a lighting load **604**. The control device **600** may include a controllably conductive device **610** coupled in series electrical connection between the AC power source **602** and the lighting load **604**. The controllably conductive device **610** may control the power delivered to the lighting load. The controllably conductive device **610** may include a relay and/or a bidirectional semiconductor switch, such as, for example, a triac, a field-effect transistor (FET) in a rectifier bridge, two FETs in anti-series connection, one or more insulated-gate bipolar junction transistors (IGBTs), or other suitable semiconductor switching circuit.

(62) The control device **600** may include a control circuit **614**. The control circuit **614** may include one or more of a processor (e.g., a microprocessor), a microcontroller, a programmable logic device (PLD), a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), or any suitable controller or processing device. The control circuit **614** may be operatively coupled to a control input of the controllably conductive device **610**, for example, via a gate drive circuit **612**. The control circuit **614** may be used for rendering the controllably conductive device **610** conductive or non-conductive, for example, to turn the lighting load **604** on and off and/or to control the amount of power delivered to the lighting load **604**.

(63) The control circuit **614** may receive a control signal representative of the zero-crossing points of the AC main line voltage of the AC power source **602** from a zero-crossing detector **616**. The control circuit **614** may be operable to render the controllably conductive device **610** conductive and/or non-conductive at predetermined times relative to the zero-crossing points of the AC waveform using a phase-control dimming technique. Examples of dimmers are described in greater detail in commonly-assigned U.S. Pat. No. 7,242,150, issued Jul. 10, 2007, entitled DIMMER HAVING A POWER SUPPLY MONITORING CIRCUIT; U.S. Pat. No. 7,546,473, issued Jun. 9, 2009, entitled DIMMER HAVING A MICROPROCESSOR-CONTROLLED POWER SUPPLY; and U.S. Pat. No. 8,664,881, issued Mar. 4, 2014, entitled TWO-WIRE DIMMER SWITCH FOR LOW-POWER LOADS, the entire disclosures of which are hereby incorporated by reference.

(64) The control device **600** may include a memory **618**. The memory **618** may be communicatively coupled to the control circuit **614** for the storage and/or retrieval of, for example, operational settings, such as, lighting presets and associated preset light intensities. The memory **618** may be implemented as an external integrated circuit (IC) or as an internal circuit of the control circuit **614**. The control device **600** may include a power supply **620**. The power supply **620** may generate a direct-current (DC) supply voltage V_{subCC} for powering the control circuit **614** and the other low-voltage circuitry of the control device **600**. The power supply **620** may be coupled in parallel with the controllably conductive device **610**. The power supply **620** may be operable to conduct a charging current through the lighting load **804** to generate the DC supply

voltage V.sub.CC.

(65) The control circuit **614** may be responsive to inputs received from one or more of the actuators **630** and/or a rotational sensing circuit **640**. The control circuit **614** may control the controllably conductive device **610** to turn the lighting load **604** on and off, adjust the intensity of the lighting load, and/or adjust the color of the lighting load in response to the inputs received via the actuators **630** and/or the rotational position sensing circuit **640**. The actuators **630** may include a button or switch (e.g., a mechanical button or switch, or an imitation thereof) such as those described in association with the actuation portion **216** of the control device **200** and/or the actuation portion **334** of the remote control device **310**. The actuators **630** may be configured to send respective input signals to the control circuit **614** in response to actuations of the actuators **630**.

(66) The rotary position sensing circuit **640** may be configured to translate a force applied to a rotating mechanism (e.g., such as the rotating portion **214** of the control device **200** and/or the rotating portion **332** of the remote control device **310**) into an input signal and provide the input signal to the control circuit **614**. The rotational position sensing circuit **640** may include, for example, one or more magnetic sensors (such as Hall-effect sensors (HES), tunneling magnetoresistance (TMR) sensors, anisotropic magnetoresistance (AMR) sensors, giant magnetoresistance (GMR) sensors, reed switches, or other mechanical magnetic sensors), a mechanical encoder, and/or an optical encoder.

(67) The control device **600** may comprise a communication circuit **622**. The communication circuit **622** may comprise a wireless communication circuit, for example, a radio-frequency (RF) transceiver coupled to an antenna for transmitting and/or receiving RF signals, an RF transmitter for transmitting RF signals, an RF receiver for receiving RF signals, or an infrared (IR) transmitter and/or receiver for transmitting and/or receiving IR signals. The communication circuit **622** may also comprise a wired communication circuit configured to be coupled to a wired control link, for example, a digital communication link and/or an analog control link, such as a 0-10V control link or a pulse-width modulated (PWM) control link. In addition, the communication circuit **622** may be coupled to the electrical wiring between the control device **600** and the lighting load **604** and may be configured to transmit a control signal to the lighting load **604** via the electrical wiring using, for example, a power-line carrier (PLC) communication technique.

(68) The communication circuit **622** may be configured to transmit a control signal that includes the control data (e.g., a digital message) generated by the control circuit **614** to the lighting load **604**. As described herein, the control data may be generated in response to a user input to adjust one or more operational aspects of the lighting load **604**. The control data may include a command and/or identification information (e.g., such as a unique identifier) associated with the control device **600**. In addition to or in lieu of transmitting the control signal to the lighting load **604**, the communication circuit **622** may be controlled to transmit the control signal to a central controller of the lighting control system.

(69) The control circuit **614** may be configured to turn the lighting load on and off by rendering the controllably conductive device **610** conductive and non-conductive in response to an actuation of one of the actuators **630**. The control circuit **614** may be configured to transmit digital messages to the lighting load **604** via the communication circuit **622** for adjusting the intensity of the lighting load in response to rotations of the rotating mechanism determined from the rotational sensing circuit **640**. In addition, the control circuit **614** may be configured to control the controllably conductive device **610** using the phase control technique to adjust the intensity of the lighting load in response to rotations of the rotating mechanism determined from the rotational sensing circuit **640**. The control circuit **614** may be configured to transmit digital messages to the lighting load **604** via the communication circuit **622** for adjusting the color of the lighting load in response to rotations of the rotating mechanism while one of the actuators **630** is being actuated.

(70) The control circuit **614** may be configured to illuminate one or more light sources, e.g., LEDs **650**, to provide feedback of a status of the lighting load **604**, to indicate a status of the control

device **600**, and/or to assist with a control operation (e.g., to provide a color gradient for controlling the color of the lighting load **604**, etc.). The LEDs **650** may be configured to illuminate one or more visual indicators, such as a light bar (e.g., the light bar **218** and/or the light bar **336**), to serve as indicators of various conditions. When the rotating mechanism is being rotated to adjust the intensity of the lighting load **604**, the control circuit **614** may control the LEDs **650** to illuminate the light bar (e.g., illuminated in a single color, such as white) to display feedback information regarding the present intensity of the lighting load **604**. When the rotating mechanism is being rotated while one of the actuators **630** is being actuated in order to adjust the color of the lighting load **604**, the control circuit **614** may control the LEDs **650** to illuminate the light bar with one or more colors to provide feedback of the present color of the lighting load **604**.

(71) FIG. 7 shows a simplified flowchart of an example control procedure **700** that may be executed by a control circuit of a control device (e.g., the control circuit **530** of the control device **500** and/or the control circuit **614** of the control device **600**) for controlling multiple characteristics of an electrical load, such as a lighting load. The control procedure **700** may begin at **710** when rotation of a rotating mechanism (e.g., the rotating portion **214** of the control device **200** and/or the rotating portion **332** of the remote control device **310**) is detected via a rotational sensing circuit (e.g., the rotational sensing circuit **534** and/or the rotational sensing circuit **640**). In response to the rotation, the control circuit may determine whether the front surface of an actuation mechanism (e.g., the actuation portion **216** and/or the actuation portion **334**) is in a first plane along an axis perpendicular to a base portion of the control device. The determination may be made, for example, based on the state of a switch (e.g., whether the switch is actuated, whether the switch is open or closed, etc.) included in the control device. The determination may also be made via a sensor (e.g., a light emitting diode (LED) sensor), a photosensitive device, an optocoupler that comprises an infra-red (IR) light emitting diode (LED) and a photodiode, an inductive sensor, a hall-effect sensor circuit, a manually operated switch, and/or the like, as described above.

(72) If the control circuit determines that the front surface of the rotating mechanism is presently in the first plane at **712**, the control circuit may adjust the intensity of the lighting load (e.g., by directly controlling a controllably conductive device and/or by transmitting one or more digital messages) at **714**, before the control procedure **700** exits. If the control circuit determines that the front surface of the actuation mechanism is presently not in the first plane at **712** while the rotating mechanism is being rotated, the control circuit may adjust the color of the lighting load (e.g., by directly controlling a controllably conductive device and/or by transmitting one or more digital messages) at **716**, before the control procedure **700** exits.

(73) Although described with reference to color and intensity, the control circuit (e.g., via the control procedure **700**) may generate control signals for adjusting any type of characteristic of an electrical load in response to a rotation of the rotating mechanism when the rotating mechanism is in a particular plane. For example, the characteristics may be any of intensity, color (e.g., color temperature), volume, music selection, HVAC mode (e.g., AC on/off, heat on/off, temperature, fan speed, etc.), ceiling fan speed, relative height/location of a motorized window treatment, or any of adjustable characteristics of the electrical loads described herein. Further, although described with reference to controlling a single electrical load, the control circuit may be configured to control a characteristic of one or more electrical load in response to a rotation of the rotating mechanism in the first plane, and another potentially different characteristic of one or more potentially different electrical loads in response to a rotation of the rotating mechanism in the second plane (e.g., and a third characteristic of one or more potentially different electrical loads in response to a rotation of the rotating mechanism in the third plane, etc.).

Claims

1. A control device for controlling a lighting load in a load control system, the control device comprising: a base portion; a rotating portion rotatable with respect to the base portion; an actuation portion having a front surface, the front surface of the actuation portion configured to be in a first plane along an axis perpendicular to the base portion and in a second plane parallel to the first plane, wherein the second plane is located closer to the base portion than the first plane, and the front surface of the actuation portion is configured to be moved from the first plane to the second plane when the rotating portion is pushed in towards the base portion; and a control circuit configured to generate first control data for changing one of color or intensity of the lighting load in response to rotations of the rotating portion when the front surface of the actuation portion is located in the first plane, the control circuit further configured to generate second control data for changing the other of color or intensity of the lighting load in response to rotations of the rotating portion when the front surface of the actuation portion is in the second plane.
2. The control device of claim 1, wherein the base portion is configured to be mounted over an actuator of a mechanical switch that controls power delivered to the lighting load.
3. The control device of claim 1, wherein the base portion is configured to be mounted to an electrical wallbox.
4. The control device of claim 3, further comprising: a load control circuit adapted to be electrically coupled in series between an AC power source and the lighting load for controlling power delivered to the lighting load.
5. The control device of claim 1, further comprising: an internal switch; wherein the actuation portion is configured to not actuate the internal switch when the front surface of the actuation portion is in the first plane, and configured to actuate the internal switch when the front surface of the actuation portion is moved into the second plane.
6. The control device of claim 5, wherein the internal switch comprises a sensor, an light emitting diode (LED) sensor, a photosensitive device, an optocoupler that comprises an infra-red (IR) LED and a photodiode, an inductive sensor, a hall-effect sensor circuit, or a mechanically operated switch.
7. The control device of claim 1, further comprising: one or more visual indicators configured to be illuminated by one or more light sources; wherein, when the rotating portion is being rotated to adjust the intensity of the lighting load, the control circuit is configured to illuminate the one or more visual indicators to display feedback information regarding a present intensity of the lighting load; and wherein, when the rotating portion is being rotated to adjust the color of the lighting load, the control circuit is configured to illuminate the one or more visual indicators to display feedback information regarding a present color of the lighting load.
8. The control device of claim 1, wherein the actuation portion is separate from the rotating portion and received in a central opening of the rotating portion.
9. The control device of claim 1, further comprising: a communication circuit configured to transmit a first control signal associated with the first control data and a second control signal associated with the second control data.
10. A control device for controlling a lighting load in a load control system, the control device comprising: a base portion configured to be mounted over a mechanical switch that controls power delivered to the lighting load; an internal switch; a rotating portion rotatable with respect to the base portion, the rotating portion comprising a front area, the rotating portion configured to not actuate the internal switch when the front area of the rotating portion is located in a first plane and to actuate the internal switch when the front area of the rotating portion is moved into a second plane parallel to the first plane; and a control circuit configured to generate first control data for changing one of color or intensity of the lighting load in response to rotations of the rotating portion when the front surface of the actuation portion is located in the first plane and the internal switch is not actuated, the control circuit further configured to generate second control data for

changing the other of color or intensity of the lighting load in response to rotations of the rotating portion when the front surface of the actuation portion is in the second plane and the internal switch is actuated.

11. The control device of claim 10, further comprising: an actuation portion comprising a front surface that defines the front area of the rotating portion.

12. The control device of claim 11, wherein the actuation portion is attached to the rotating portion.

13. The control device of claim 11, wherein the actuation portion is separate from the rotating portion and received in a central opening of the rotating portion.

14. The control device of claim 10, wherein the second plane is located closer to the base portion than the first plane, and the front area of the rotating portion is capable of being moved from the first plane to the second plane when the rotating portion is pushed in towards the base portion.

15. The control device of claim 10, wherein the second plane is located farther away from the base portion than the first plane, and the front area of the rotating portion is capable of being moved from the first plane to the second plane when the rotating portion is pulled away from the base portion.

16. The control device of claim 1, wherein the actuation portion is configured to not actuate the internal switch when the front surface of the actuation portion is in the first plane, and configured to actuate the internal switch when the front surface of the actuation portion is moved into the second plane.

17. The control device of claim 16, wherein the internal switch comprises a sensor, an light emitting diode (LED) sensor, a photosensitive device, an optocoupler that comprises an infra-red (IR) LED and a photodiode, an inductive sensor, a hall-effect sensor circuit, or a mechanically operated switch.

18. A method for controlling a lighting load in a load control system using a control device, the method comprising: detecting rotation of a rotating portion of the control device; determining whether a front surface of an actuation mechanism of the control device is in a first plane or a second plane along an axis perpendicular to a base portion of the control device; wherein, when the front surface of the actuation mechanism is in the first plane, the method further comprises adjusting one of color or intensity of the lighting load; and wherein, when the front surface of the actuation mechanism is in the second plane, the method further comprises adjusting the other of color or intensity of the lighting load.

19. The method of claim 18, further comprising: controlling a controllably conductive device of the control device to adjust one of the color or intensity of the lighting load when the front surface of the actuation mechanism is in the first plane; or controlling the controllably conductive device of the control device to adjust the other of the color or intensity of the lighting load when the front surface of the actuation mechanism is in the second plane.

20. The method of claim 18, further comprising: transmitting a first digital message using a communication circuit of the control device to adjust one of the color or intensity of the lighting load when the front surface of the actuation mechanism is in the first plane; and transmitting a second digital message using the communication circuit to adjust the other of the color or intensity of the lighting load when the front surface of the actuation mechanism is in the second plane.

21. The method of claim 18, further comprising: detecting that an internal switch of the control device is not actuated to determine that the front surface of the actuation mechanism is in the first plane; and detecting that the internal switch is actuated to determine that the front surface of the actuation mechanism is in the second plane.

22. The method of claim 21, wherein the internal switch comprises a sensor, an light emitting diode (LED) sensor, a photosensitive device, an optocoupler that comprises an infra-red (IR) LED and a photodiode, an inductive sensor, a hall-effect sensor circuit, or a mechanically operated switch.

23. The method of claim 22, wherein the second plane is located closer to the base portion than the

first plane, and the front area of the rotating portion is capable of being moved from the first plane to the second plane when the rotating portion is pushed in towards the base portion.
